

# Supporting Information for ‘Machine Learning and Computer Simulation Disentangle the Fuzzy Inhibitor Binding by Hsp90’

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## Detailed Simulation Methods and Analysis

### Unbiased molecular dynamics simulations

Five sets of all-atom, unbiased molecular dynamics (MD) simulations were conducted for the N-terminal domain of human Hsp90 (N-Hsp90, residue 9-236). An aggregated total of 70  $\mu$ s of simulation was performed (Table S1).

**i) Unbiased binding kinetics simulations:** In the first set, unbiased binding kinetic simulations were performed to investigate the binding of Geldanamycin (GDM) to the apo form of N-Hsp90. GDM molecules, initially placed in the solvent phase, were allowed to explore the environment around the apo N-Hsp90 in search of the native binding pocket. The starting configuration of human Hsp90 $\alpha$  was taken from previously solved crystal structures of the apo form (PDB ID: 1YES, loop-out conformation, figure 2A). The all-atom MD system comprised N-Hsp90 with residues protonated at neutral pH, neutralized with 180 mM KCl, and solvated in a TIP3P water box with a 10 Å buffer around the protein. The CHARMM36<sup>1</sup>

force field was used to parameterize the protein. For the binding simulations, four GDM molecules, corresponding to a concentration of 10 mM, were randomly placed in the vicinity of the apo protein. The force field parameters for GDM were derived using GAAMP.,<sup>2</sup> which enabled us to employ optimized partial charges as well as the optimized bonded parameters. All simulations were preceded by 250 ps of NVT and NPT equilibrations, employing the v-rescale thermostat<sup>3</sup> (310.15 K) and the Parrinello-Rahman<sup>4</sup> barostat. The simulations were executed using GROMACS 20xx,<sup>5</sup> with non-bonded interactions calculated via Particle Mesh Ewald (PME)<sup>6</sup> summation and the Verlet<sup>7</sup> scheme within a 1.2 nm cutoff. Hydrogen bonds were constrained using the LINCS<sup>8</sup> and SETTLE<sup>9</sup> algorithms.

**ii) Intermediate state simulations** To comprehensively sample the ligand-binding phase space for subsequent Markov state model (MSM) development (described later), multiple configurations corresponding to intermediate states were selected from the long binding trajectories. 20 shorter simulations (100 ns each) were then adaptively initiated from these intermediate states.

**iii) and iv) Comparative simulations of apo and bound states:** For a comparative assessment of the conformational heterogeneity of Hsp90 in its apo and native-bound forms, additional simulations were performed starting from crystal structures corresponding to the apo form (PDB IDs: 1YES, 1YER) and the bound form (PDB ID: 1YET).<sup>10</sup>

**v) Supplemental adaptive sampling:** In addition to the simulations described above, further efforts were undertaken to adaptively sample sparsely populated conformations and thereby enrich the interconversion between distinct metastable states. To this end, we first performed an extensive set of  $\tau$ -RAMD simulations, followed by multiple rounds of targeted seeding molecular dynamics (MD) simulations (described in detail in the subsequent subsections). These simulations yielded a total of 82 representative conformations. Each of these conformations were sampled via 100 ns of MD simulations.

**Table S1:** Number of replicates and time scales for unbiased simulations. Unbiased binding simulations were (6.4,4,2,0.5,2,2,2)  $\mu s$  in lengths. Apo simulations were  $5 \times 4 + 3.26 \mu s$ , while bound, intermediate and adaptive samplings were 4  $\mu s$ , 100, 100 ns of lengths respectively.

| simulation                   | number of replicates | total simulation time ( $\mu s$ ) |
|------------------------------|----------------------|-----------------------------------|
| Unbiased binding simulations | 8                    | 20.97                             |
| Intermediate sampling        | 20                   | 2                                 |
| Apo state                    | 6                    | 23.26                             |
| Bound state                  | 4                    | 16                                |
| Adaptive sampling            | 82                   | 8.2                               |

## Development of Markov State Models

Three independent Markov state models (MSMs)<sup>11,12</sup> were constructed using different sets of unbiased MD data to investigate: (i) the GDM binding process to its native pocket in Hsp90, and (ii) and (iii) the loop dynamics near the L4 region of Hsp90 in both apo and bound states.

To construct the Markov State Model (MSM) of the inhibitor binding process, we employed a comprehensive set of unbiased and adaptively sampled molecular dynamics (MD) simulations. Specifically, the MSM was built using the unbiased binding simulations presented in Fig. 2A, along with simulations initiated from the bound and intermediate states defined in Table 1. In addition, adaptive sampling simulations were performed using configurations extracted from  $\tau$ -RAMD simulations and targeted seeding simulations (see subsequent subsections for methodological details). In total, 108 trajectories amounting to an aggregate simulation time of 38.5  $\mu s$  were utilized for MSM construction of GDM binding process to Hsp90.

For state space discretization, we selected four collective variables (CVs): (1) the center-of-geometry distance between GDM and the binding pocket residues (residues 51, 54, 58, 98, 102, 106, 107, 135, and 138); (2) the minimum heavy-atom distance between GDM and the N-terminal domain of Hsp90; (3) the RMSD of the GDM molecule relative to its bound conformation; and (4) the orientation of GDM with respect to its bound pose (Fig. S1). The orientation was quantified as the angle between the GDM vector defined by atoms C6  $\rightarrow$  C20

and a protein vector defined by the  $C_\alpha$  atoms of residues V150 and W162. These CVs were used to discretize the simulated trajectories as the initial step in constructing the MSM (Table S2). Notably, an attempt to use the GDM-protein contact matrix as a clustering metric did not successfully resolve the states.

For the MSM characterizing the loop dynamics involving the L4 region (residues 106-115), minimum distances between each loop residue and every third residue of the rest of the protein (resulting in a 630-dimensional CV array) were considered. To reduce the dimensionality of the input CVs for the loop dynamics MSM, time-structured independent component analysis (TICA)<sup>13</sup> was employed to extract the two slowest motions. These were subsequently clustered into microstates using the k-means algorithm.

Table S2 lists the relevant hyperparameters used in the respective MSMs. The final MSMs were built over a range of lag times and were coarse-grained into 2-7 states using the PCCA (Perron Cluster Cluster Analysis)<sup>14</sup> algorithm (ITS plots in Fig. 2, S12).<sup>15</sup> The final settings were chosen based on the convergence of PCCA states across all lag times (Table S2). Pathways and associated transition timescales were calculated using transition path theory.<sup>16</sup> The MSMs were constructed and analyzed using the Python libraries `pyemma`.<sup>17</sup>

**Table S2:** Hyperparameters used in markov state models. The value in brackets of data size represents the number of trajectories.

| MSM                       | data size ( $\mu s$ ) | tica  | clusters | lag time (ns) | pcca states |
|---------------------------|-----------------------|-------|----------|---------------|-------------|
| binding process           | 38.5 (108)            | 20 ns | 200      | 0.2           | 3           |
| loop dynamics - Apo (6)   | 23.26                 | 5 ns  | 50       | 0.2           | 6           |
| loop dynamics - Bound (4) | 16                    | 5 ns  | 50       | 0.2           | 4           |

## $\tau$ -RAMD simulations

$\tau$ -RAMD simulations<sup>18</sup> were employed to generate initial seeds for subsequent adaptive sampling simulations, thereby enhancing the statistical robustness of the MSM. These simulations were initiated from both the bound and intermediate states. For the bound state,



32 starting structures were extracted from a 16  $\mu$ s trajectory, sampled at 500 ns intervals to capture conformational heterogeneity across time. For the intermediate state, 20 starting structures were selected from unbiased binding simulations at uniformly spaced time intervals. For each starting conformation, whether from the bound or intermediate ensemble, 15 independent  $\tau$ -RAMD replicates were performed with different random seeds.

In these simulations, a random expulsion force of 360 kJ/mol was applied to the GDM molecule to induce unbinding. The direction of the random force was updated every 50 integration steps, conditioned on a minimum displacement of 0.0025 nm. Each simulation was terminated when the center-of-mass distance between the N atom of GDM and the C $_{\alpha}$  atom of residue V150 (N-Hsp90) exceeded 4.0 nm, indicating successful unbinding.

Additionally, the  $\tau$ -RAMD trajectories were analyzed to estimate the relative residence times of the bound and intermediate states, following the procedure described in ref.<sup>19</sup> For each starting structure, unbinding times from 15 replicates were recorded. These were bootstrapped over 50,000 rounds by randomly selecting 12 replicates per round. The final residence time was reported as the mean and standard deviation of the median unbinding time obtained from the bootstrapped samples. To ensure adequate sampling, the empirical cumulative distribution function (CDF) of the unbinding times was compared with the theoretical Poisson CDF,  $P(n \geq 1) = 1 - \exp(-t/\tau)$ , using the Kolmogorov-Smirnov (KS) statistic. The  $\tau$ -RAMD simulations were performed via its Gromacs-2020.5 implementation.<sup>20</sup>

## Targeted simulations for seeding adaptive sampling

Targeted sampling simulations were performed to generate starting conformations for two distinct purposes: (i) additional adaptive sampling of the binding process, and (ii) infrequent metadynamics simulations of the GDM-bound excited state, whereby crystal structures are not available.

First, targeted simulations were employed to generate intermediate conformations that bridge key transitions: (a) between the bound state and the intermediate state observed in

unbiased binding simulations, and (b) between distinct intermediate conformations identified in the two binding trajectories (Fig. 2A). In round 1, five independent simulations of 3 ns each were initiated from the designated starting state. For each subsequent round, starting structures were selected based on their proximity to the target state, quantified by the RMSD of the GDM molecule. To facilitate rare transitions, simulations were conducted at an elevated temperature of 500 K. The conformation of N-Hsp90 was position-restrained to avoid structural distortions under high-temperature conditions. Iterative rounds of simulation were continued until the target state was achieved within a predefined RMSD threshold (in Å). The desired transitions were successfully obtained within 7-20 rounds of simulations. In total, 32 seeding configurations were generated for further adaptive sampling at normal temperatures.

In addition, targeted sampling simulations were conducted to generate a GDM-bound conformation of N-Hsp90 in its *excited* or low-population state.<sup>21</sup> Notably, this excited conformation has been experimentally resolved only in the apo state (PDB: 8B7J) and in complex with a non-GDM inhibitor (PDB: 7UR3). Direct placement of GDM into these structures led to severe steric clashes (Fig. S8), rendering standard docking or relaxation approaches ineffective. To overcome this, we initiated targeted MD simulations from the GDM-bound ground-state structure (in the loop-out conformation, as defined above), steering the system toward the excited-state geometries of PDB 8B7J and 7UR3.

In round 1, four simulations of 3 ns each were launched. For subsequent rounds, starting conformations were selected based on the RMSD of residues 106-136 relative to the excited-state target structures. Simulations were again performed at 500 K to overcome the high energy barriers, consistent with the millisecond-scale timescales associated with these transitions.<sup>21</sup> All N-Hsp90 residues were position-restrained, except residues 106-136, to allow for localized conformational adaptation. Within this segment, helicity of the  $\alpha 5$  helix was preserved using a flat-bottom umbrella potential applied to hydrogen bonds involving residue pairs 115-119, 116-120, 117-121, 118-122, and 119-123, with a lower-bound distance of 3.1 Å

and a restraining force constant of 10,000 kJ/mol. Ultimately, the GDM-bound excited-state conformations closely resembling PDB 8B7J and 7UR3 were successfully achieved in 110 and 130 simulation rounds, respectively.

## Funnel metadynamics simulations

The binding of GDM to N-Hsp90 was independently modeled using the funnel metadynamics simulation approach,<sup>22,23</sup> which applies a funnel-shaped (cone + cylinder) restraining potential to confine the ligand’s conformational space in the unbound region while preserving unbiased sampling within the binding pocket. The funnel geometry for each system is illustrated in Figs. S6A and S8). The conical section encompasses the entire binding pocket, including loops L3 and L4, and helices  $\alpha 2$ ,  $\alpha 4$ ,  $\alpha 5$ , and  $\alpha 6$  (Fig. S6A). Within this region, GDM is free to explore its conformational space without bias. The cylindrical section restricts GDM’s motion in the unbound state, thus accelerating sampling without perturbing the bound region. Independent funnel metadynamics simulations were performed for three conformations of N-Hsp90: loop-out, loop-in, and excited states.

For the loop-out conformation, the starting structure was taken from the bound-state simulation based on PDB ID 1YET. The conical portion of the funnel restraint extended up to 2.0 nm from the pocket center (denoted  $Z_{cc}$ ), with a cone angle parameter  $\alpha = 0.55$ , ensuring inclusion of all relevant pocket regions. The cylindrical extension reached an additional 2.0 nm (total funnel length of 4.0 nm), with a radius  $R_{cyl} = 0.1$  nm. The funnel potential was applied with a force constant  $\kappa = 35,100$  kJ/mol/nm<sup>2</sup>, and tabulated on a  $500 \times 500$  grid. Lower and upper restraining walls along the funnel axis were placed at 0.2 nm and 3.8 nm, respectively, each with a force constant of 500,000 kJ/mol/nm<sup>2</sup>.

Well-tempered metadynamics<sup>24</sup> biasing was applied to the GDM molecule as a function of its distance from the  $C_\beta$  atom of residue V150. Gaussians with height 1.2 kJ/mol and width 0.035 nm were deposited every 500 steps, using a bias factor of 15. The bias was applied within the 0.2-3.8 nm range along the funnel axis. Additionally, an RMSD upper

wall was applied to N-Hsp90 (with a force constant of 500,000 kJ/mol/nm<sup>2</sup>), constraining it within 1 Å of the initial conformation. Eight independent funnel metadynamics simulations of 300-400 ns each were carried out, successfully capturing numerous binding events. The final binding free energy was estimated using the `sum_hills` utility from the PLUMED<sup>25</sup> plugin patched with GROMACS.

For the loop-in conformation, the starting structure was extracted from the unbiased binding simulations where the intermediate state was observed in the loop-in form. Eight independent replicates of 300-400 ns funnel metadynamics were carried out using the same funnel parameters as for the loop-out conformation.

To explore the possibility of GDM binding in excited conformations, both known excited states of N-Hsp90 (PDB IDs: 7UR3 and 8B7J) were considered. In the 7UR3 conformation,  $\alpha 5$  and the unfolded  $\alpha 6$  lie adjacent to  $\alpha 4$ , leaving a potential path to the binding pocket, similar to the ground state (Fig. S8). Consequently, eight independent funnel metadynamics simulations (300-400 ns each) were performed using the same funnel setup as above.

In contrast, the excited conformation from PDB ID 8B7J presents a different topology, where  $\alpha 5$  and the unfolded  $\alpha 6$  occlude the binding pocket by shifting toward  $\alpha 2$  (Fig. S8). Therefore, two distinct funnel restraint configurations were employed. In the first, the same funnel setup used for previous conformations was applied (4 replicates, each 300-400 ns). In the second setup, the funnel restraint was localized beneath  $\alpha 5$  and above  $\alpha 4$  and L3 (Fig. S8). Here, the conical section extended 2.5 nm with  $\alpha = 0.55$ , and the total funnel length remained 4.0 nm. The well-tempered metadynamics bias was applied based on the distance from the  $C_\alpha$  atom of residue G137, with all other biasing parameters identical to previous setups.

## Inhibitor residence time and off-rate constant estimation by infrequent metadynamics simulations

The reverse binding process and estimation of the dissociation rate constant ( $k_{\text{off}}$ ) for the GDM molecule were performed using infrequent metadynamics simulations<sup>23</sup> for both the ground and excited conformations of N-Hsp90.

For the ground state (loop-out conformation), 64 starting structures were extracted from the 16  $\mu\text{s}$  of unbiased bound-state simulations. Specifically, one frame with the smallest GDM-to-pocket distance was selected from every 250 ns segment. Among these, 16 frames were repeated to assess reproducibility, resulting in a total of 80 independent infrequent metadynamics replicates for the ground state.

For the excited state, 30 starting frames were selected from the final 30 rounds of the targeted MD simulations used to generate the GDM-bound excited conformations (as described above). These frames were obtained from both known excited conformations of N-Hsp90 (PDB IDs: 8B7J and 7UR3), resulting in a total of 60 independent replicates for the excited state unbinding simulations.

A history-dependent Gaussian bias potential  $V(s, t)$  was applied as a function of the distance between the GDM molecule and the binding pocket (see Fig. S1). For the ground state, the bias height was 1.2 kJ/mol with a width of 0.038 nm, deposited every 5,000 steps (10 ps) using a bias factor of 15. For the excited state, a higher bias height of 4.8 kJ/mol and a bias factor of 50 were employed to account for the anticipated longer residence times. In some excited-state replicates, lower bias heights (1.8 kJ/mol) and bias factors (15) were also tested to verify reproducibility.

The acceleration factor  $\alpha$  (eq. 1) was calculated based on the time-dependent accumulated bias potential until complete unbinding was observed, defined as the center-of-mass distance between the GDM molecule and the binding pocket exceeding 3.0 nm. The unbiased unbinding time as estimated as in eq2:

$$\alpha = \langle e^{\beta V_{(s,t)}} \rangle_{T_M} \quad (1)$$

$$T_u = T_M * \alpha \quad (2)$$

where,

$V_{(s,t)}$  is history dependent bias potential as a function of collective variable (s) and time (t).

$\beta = \frac{1}{k_B T}$  is inverse temperature (T),  $k_B$  is Boltzman constant.

Ultimately, a set of  $n$  unbiased unbinding times  $t^u = \{t_1^u, t_2^u, ..t_n^u\}$  were used to estimate the empirical cumulative distribution function (ECDF, eq3) and was fitted to cumulative distribution function of poisson distribution (CDF, eq4), as unbinding statistics are supposed to follow poisson distribution.<sup>26</sup> The fitted  $\tau$  represents the residence time. The goodness of fit was confirmed using two sample Kolmogorov-Smirnov test<sup>27</sup> and p-value greater than 0.001 was considered good. To fit the data, some replicates were removed from analysis. The error was estimated via 1000 bootstrapped iterations by 90% of observations to fit eq4. The  $k_{off}$  was estimated as inverse fitted residence time ( $\tau$ , eq5).

$$ECDF(t) = \frac{1}{n} \sum_i^t I(t_i^u < t) \quad (3)$$

$$P_{n \geq 1} = 1 - \exp\left(-\frac{t}{\tau}\right) \quad (4)$$

$$k_{off} = \frac{1}{\tau} \quad (5)$$

where;

$I(.)$  is indicator function which equals 1 and 0 for true and false argument respectively

$P_{n \geq 1}$  is probability of observing atleast one unbinding within time t.

# FAST adaptive sampling for unbiased inter-conversion of L4 conformations

Fluctuation Amplification of Specific Traits (FAST) adaptive sampling<sup>?</sup> was employed to enable unbiased interconversion between different conformational states of loop L4. In this approach, simulations are initiated from a defined ‘start’ structure and iteratively guided toward a ‘target’ structure across multiple rounds, without applying any external bias.

In round 1, a set of  $8 \times 10$  ns simulations was launched from the initial start conformation, generating  $8 \times 1000$  simulation frames. The selection of frames for initiating the next round was based on a composite scoring scheme involving two criteria.

Each frame was first evaluated for its similarity to the target structure, quantified by the RMSD of L4 ( $R_{target}$ ). This score was minimized and normalized to the range  $[0, 1]$  as follows:

$$\bar{R}_{target} = \frac{R_{target}^{max} - R_{target}}{R_{target}^{max} - R_{target}^{min}} \quad (6)$$

Next, each frame was scored based on its potential to explore new pathways toward the target structure. To do this, the total RMSD of L4 in a given frame relative to all other frames ( $R_{sim}$ ) was computed, and normalized such that structurally distinct frames received higher scores:

$$\bar{R}_{sim} = \frac{R_{sim} - R_{sim}^{min}}{R_{sim}^{max} - R_{sim}^{min}} \quad (7)$$

The final score for each frame was computed as a weighted linear sum of the two normalized scores:

$$score = \bar{R}_{target} + \alpha \bar{R}_{sim} \quad (8)$$

The weighting factor  $\alpha$  was adjusted across rounds to control the exploration-exploitation trade-off. In early rounds, exploration was prioritized with  $\alpha = 1$ , while in subsequent rounds,  $\alpha$  was gradually decreased to focus more on convergence toward the target:

$$\alpha = 1 - 0.02(r - 1) \tag{9}$$

In each round, eight new frames with the highest composite scores were selected as starting points for the next round of simulations ( $8 \times 10$  ns). This iterative process was continued for up to 50 rounds or until convergence toward the desired target conformation was achieved.

Using this strategy, targeted adaptive sampling was performed to model multiple transitions among the L4 loop conformations, including: loop-in to helical, loop-in to loop-out, loop-out to helical, and loop-out to loop-in. The initial (start) conformations for each transition were selected from the apo-state simulations based on their structural proximity to the respective target conformation, as quantified by L4 RMSD.

The reference structures used to define the target conformations were as follows: loop-in from PDB ID 1YER\_A, loop-out from 1YES\_A, and helical L4 from 9AUU\_A and 8AGI\_B.

## Ligand Interactions

Ligand interactions with N-Hsp90 in both intermediate and bound states were quantified by calculating the probability of contact between specific residues and the GDM ligand. A contact is defined as a minimum distance  $< 0.51$  nm between the heavy atoms of any N-Hsp90 residue and the GDM. These were converted in to probabilities by averaging over all frames from the intermediate and bound states as categorized by MSM.



## Perturbation response scanning

The propensity for interconversion between the loop-in and loop-out states was assessed using perturbation response scanning (PRS)<sup>28</sup> based on linear response theory, as implemented in the `md-task`<sup>29</sup> utility. The crystal structures of the loop-in (PDB ID: 1YER) and loop-out (PDB ID: 1YES) conformations were used as reference end states. All simulation trajectories of the apo form of Hsp90 (Table S1) were employed to construct the variance-covariance matrix. A total of 1000 perturbation forces were applied to each residue. PRS was performed for two transitions: ‘loop-in to loop-out’ and ‘loop-out to loop-in.’

## Estimation of NMR NOE distance violations

A set of 19 distance nuclear Overhauser effect (NOE) restraints measured for the ground-state R60A mutant of N-Hsp90 was utilized from previous experimental work.<sup>21</sup> Each NOE defines a distance constraint between the methyl groups of a pair of residues  $(i, j)$  (see Table S3). From the apo-state simulations, the inter-residue distances between the centers of geometry of the three methyl protons were computed for each NOE pair, denoted as  $d_{ij}^{\text{sim}}$ .

The average distance violations were then estimated as:

$$v_{ij} = \max(0, d_{ij}^{\text{sim}} - d_{ij}^{\text{NOE}}) \quad (10)$$

$$\langle v_{ij} \rangle = \frac{1}{N} \sum v_{ij} \quad (11)$$

whereby,  $d_{ij}^{\text{NOE}} = \langle d_{ij}^{\text{NMR}} \rangle + 2\sigma(d_{ij}^{\text{NMR}})$  is used from the previous work, which was estimated from the distances measured from 20 NMR model structures.

**Table S3:** Caption

| Residue <sub><i>i</i></sub> | Residue <sub><i>j</i></sub> | $d_{ij}^{NOE}$ |
|-----------------------------|-----------------------------|----------------|
| 24-HB                       | 119-HE                      | 11.87          |
| 26-HD1                      | 115-HG2                     | 5.54           |
| 26-HD1                      | 119-HE                      | 8.59           |
| 29-HD2                      | 115-HG2                     | 6.63           |
| 29-HD2                      | 119-HE                      | 6.16           |
| 29-HD2                      | 122-HD2                     | 8.89           |
| 32-HD2                      | 119-HE                      | 4.52           |
| 32-HD2                      | 122-HD2                     | 6.24           |
| 32-HD2                      | 131-HD1                     | 9.65           |
| 36-HG2                      | 122-HD2                     | 8.53           |
| 36-HG2                      | 131-HD1                     | 11.65          |
| 43-HD1                      | 122-HD2                     | 9.42           |
| 111-HB                      | 119-HE                      | 10.55          |
| 115-HG2                     | 136-HG2                     | 6.64           |
| 119-HE                      | 136-HG2                     | 8.76           |
| 122-HD2                     | 136-HG2                     | 8.31           |
| 122-HD2                     | 143-HD2                     | 7.98           |
| 32-HD2                      | 115-HG2                     | 10.72          |
| 32-HD2                      | 126-HB                      | 8.53           |

## Comparison of simulation and experimental conformational heterogeneity

A total of 422 crystal and NMR structures of human Hsp90 (UniProt ID: P07900),<sup>30</sup> publicly available in the RCSB Protein Data Bank, were analyzed in this study. With the exception of six full-length structures, all entries correspond to the N-terminal domain of Hsp90 (N-Hsp90). Our analysis focused specifically on residues 25-210.

Structures lacking backbone atoms within this residue range were excluded from further consideration. For entries containing multiple chains within the asymmetric unit or multiple NMR models, each chain or model was treated as an independent structural instance. Notably, three NMR structures (PDB IDs: 8JRA, 8JR6, and 8JR7) displayed significant structural deviation from the rest of the dataset, and no associated peer-reviewed publica-

tions could be identified for these entries. Consequently, these structures were excluded from the final analysis.

After applying these criteria, a curated dataset of 509 N-Hsp90 chains was compiled for subsequent investigation.

## Training of Machine learnt classifier

Two sets of input features were calculated from all simulations of apo and bound states (Table S1): (i) native contacts, defined as minimum distances between all residue pairs with  $C_\alpha - C_\alpha$  distances within 10 Å in the starting structure, and (ii) loop-4 distances, defined as minimum distances between each loop-4 residue (106-115) and every third residue of the rest of the protein (also used in MSM). The two feature sets contained 1194 and 630 dimensions, respectively. We opted to employ random forest as machine learning classifier. Random forest algorithms<sup>31,32</sup> can be used to measure the importance scores of each feature in a supervised classification task. Two random forest (RF) classifiers were trained on each feature set.

For the first classifier, features from only apo simulation trajectories were labeled according to six MSM states, which were further coarse-grained into three states: loop-out (0), loop-in (1), and loop (2). The second classifier used features from both apo and bound simulations, labeled as apo (0) and bound (1). Each RF model consisted of 1000 estimators, using Gini impurity as a measure of leaf purity. Fifty independent cross-validated RF models were trained by randomly selecting 70% of the training data, with the remaining 30% used for validation based on accuracy (Eq. 12) and F1 scores (Eq. 13). The final output of successfully trained classifiers was the feature importance scores ( $F_{imp}$ , Eqs. 14-15).

$$accuracy = \frac{TP + TN}{TP + FP + TN + FN} \quad (12)$$

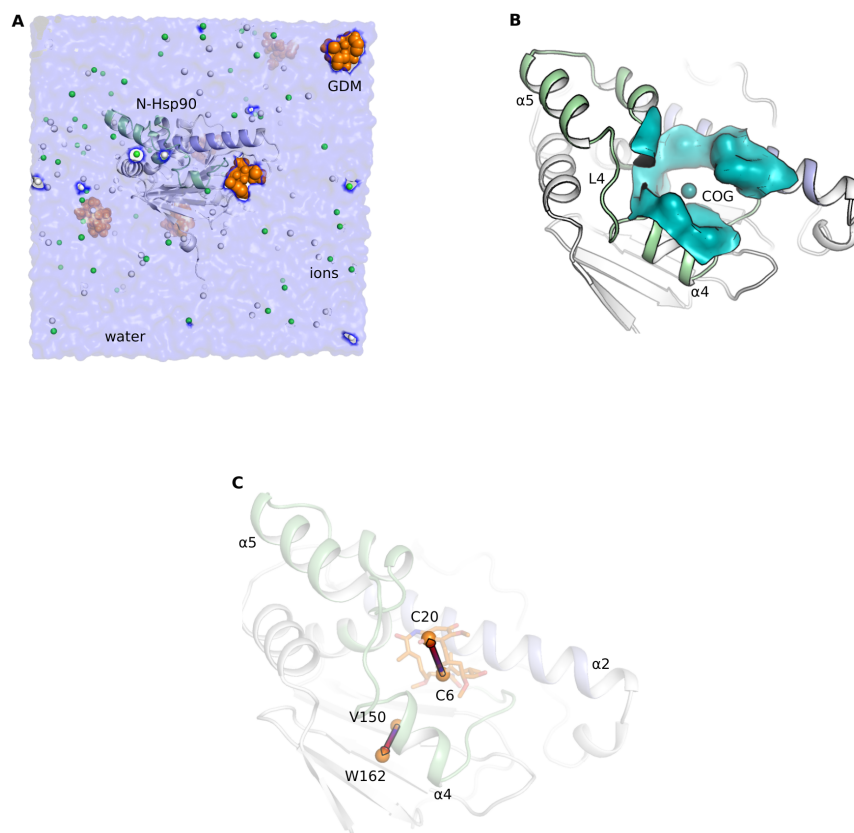
$$f_1 = \min_i \left( \frac{2 \times C_{ii}}{\sum^k C_{ik} + \sum^k C_{ki}} \right) \quad (13)$$

where, TP, FP, TN, and FN are true positive, false positive, true negative and false negative components of a k (k=i) dimensional confusion matrix estimated on test data.

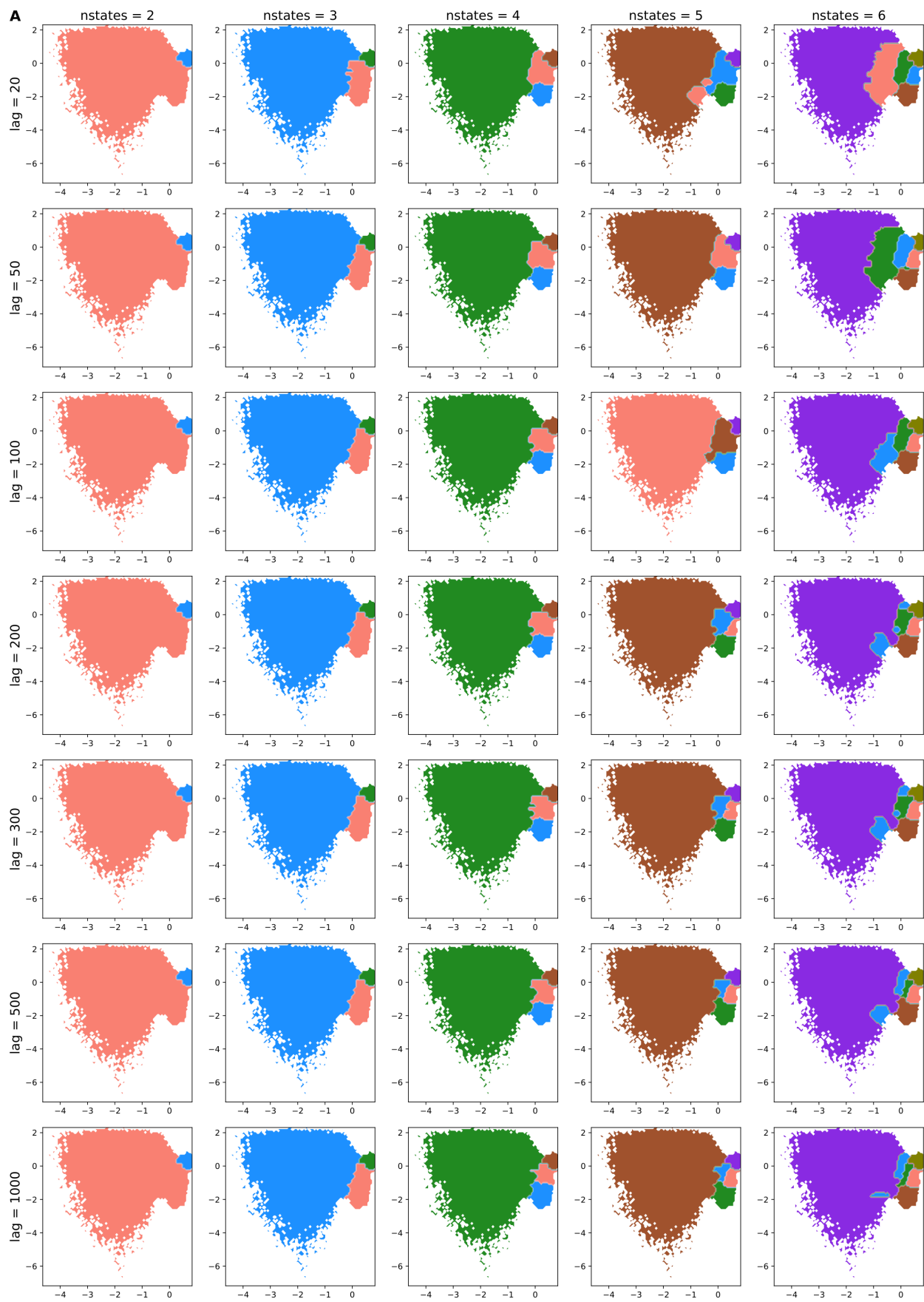
$$\Delta G_s = G_{parent} - \left( \frac{n_L}{n_{parent}} G_L + \frac{n_R}{n_{parent}} G_R \right) \quad (14)$$

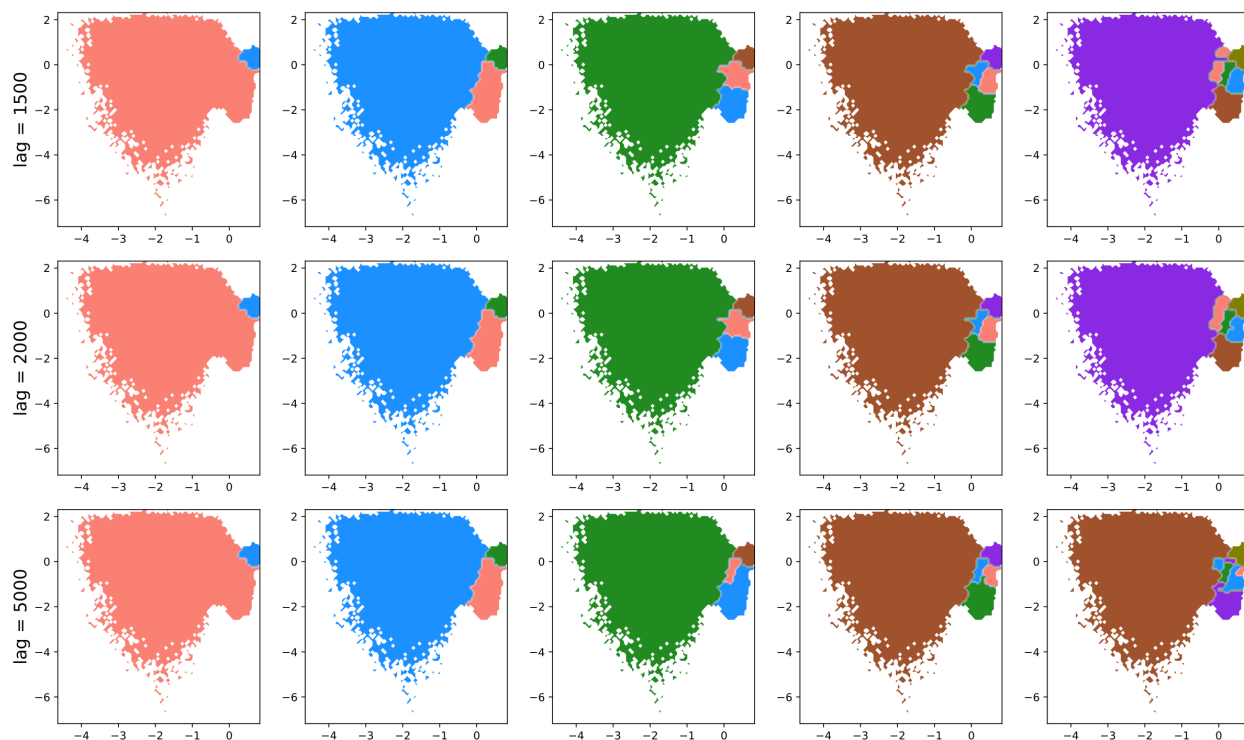
$$F_{imp}^i = \frac{1}{T} \sum_{t=1}^T \sum_s^{splits_i} \frac{n_{parent}}{n_{tree}} \Delta G_s \quad (15)$$

given that an arbitrary parent node with  $G_{parent}$  impurity and  $n_{parent}$  datapoints is split into child nodes having  $G_R$  and  $G_L$  impurities and  $n_R$  and  $n_L$  datapoints.  $F_{imp}$  is weighted mean (by datapoints in tree ( $n_{tree}$ ) and  $n_{parent}$ ) of information gain ( $\Delta G_s$ ) over all splits in all estimators ( $T$ ).



**Figure S1:** (A) The simulation setup for unbiased binding simulations, whereby apo N-Hsp90 is placed in center, surrounded by four GDM molecules and filled with water solvent. (B) The visual depiction of pocket residues (teal surface) used for estimation of center of geometry distance between binding pocket and GDM molecules. The residues include N51, D54, K58, M98, D102, N106, L107, G135 and F138. This is only for distance measurement. (C) The visual depiction of angle between protein vector ( $V150CA \rightarrow W162CA$ ) and ligand vector ( $C6 \rightarrow C20$ ), used to estimate the relative orientation of ligand molecule.





**B**

| State            | MSM Population (%) | Estimated populations (%)<br>via MSM + Metadynamics |
|------------------|--------------------|-----------------------------------------------------|
| Unbound (U)      | $28.65 \pm 1.16$   | 0.000005                                            |
| Intermediate (I) | $17.99 \pm 0.69$   |                                                     |
| Bound (B)        | $53.36 \pm 1.49$   | 99.999995                                           |

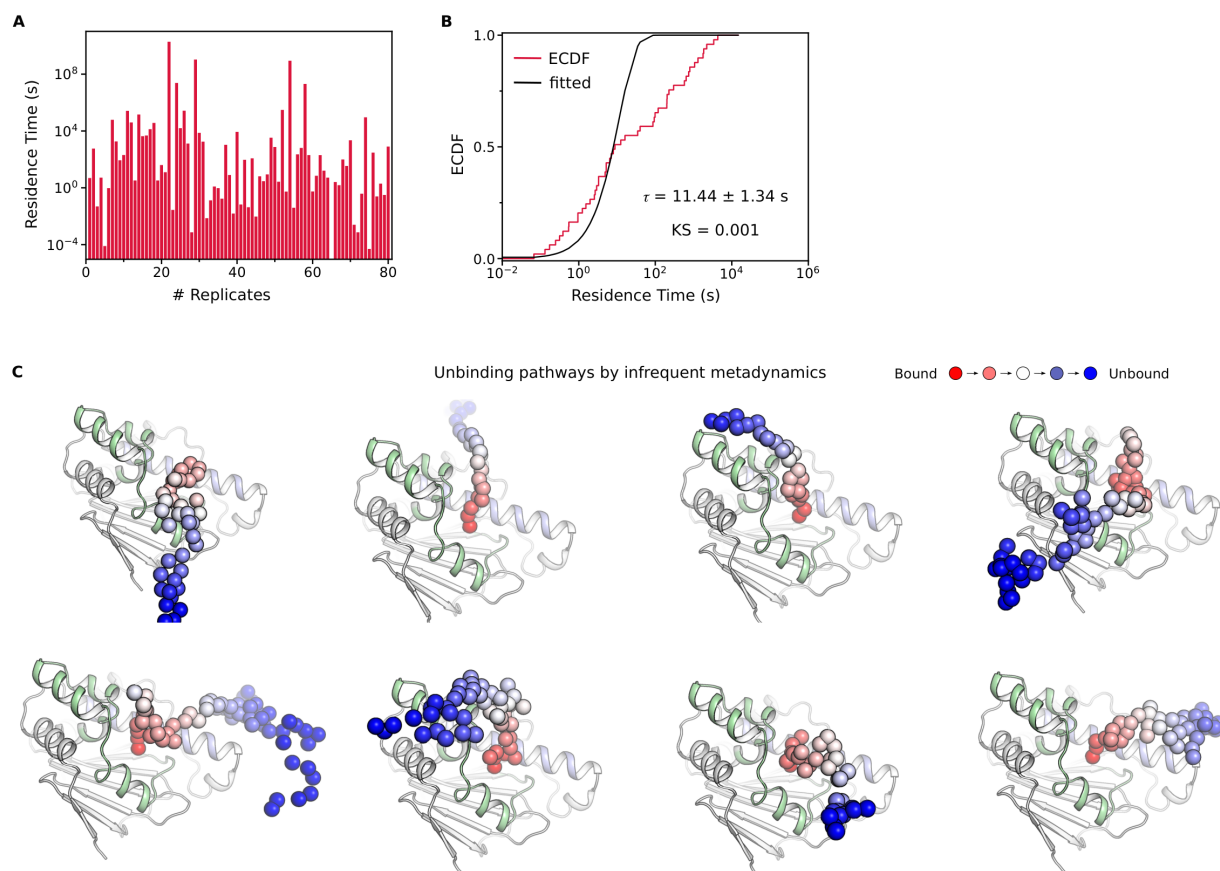
**C**

|                  | Unbound (U)                   | Intermediate (I)              | Bound (B)                     |
|------------------|-------------------------------|-------------------------------|-------------------------------|
| Unbound (U)      | $1.041 \pm 0.021 \times 10^6$ | $6.211 \pm 3.156 \times 10^3$ | $6.050 \pm 0.703 \times 10^1$ |
| Intermediate (I) | $6.484 \pm 3.129 \times 10^3$ | $8.858 \pm 0.278 \times 10^5$ | $4.881 \pm 0.498 \times 10^2$ |
| Bound (B)        | $4.650 \pm 0.537 \times 10^1$ | $4.148 \pm 0.445 \times 10^2$ | $1.834 \pm 0.017 \times 10^6$ |

**Figure S2:** (A) The MSM statemaps of GDM binding mechanism with varying lag times (number of 10 ps timesteps) and number of metastable states. The two state process outputs only unbound (including simulation observed intermediate) and bound states, compared to unbound, intermediate and bound states for 3 state MSM. Classification to 4 state MSM starts dissecting the ...

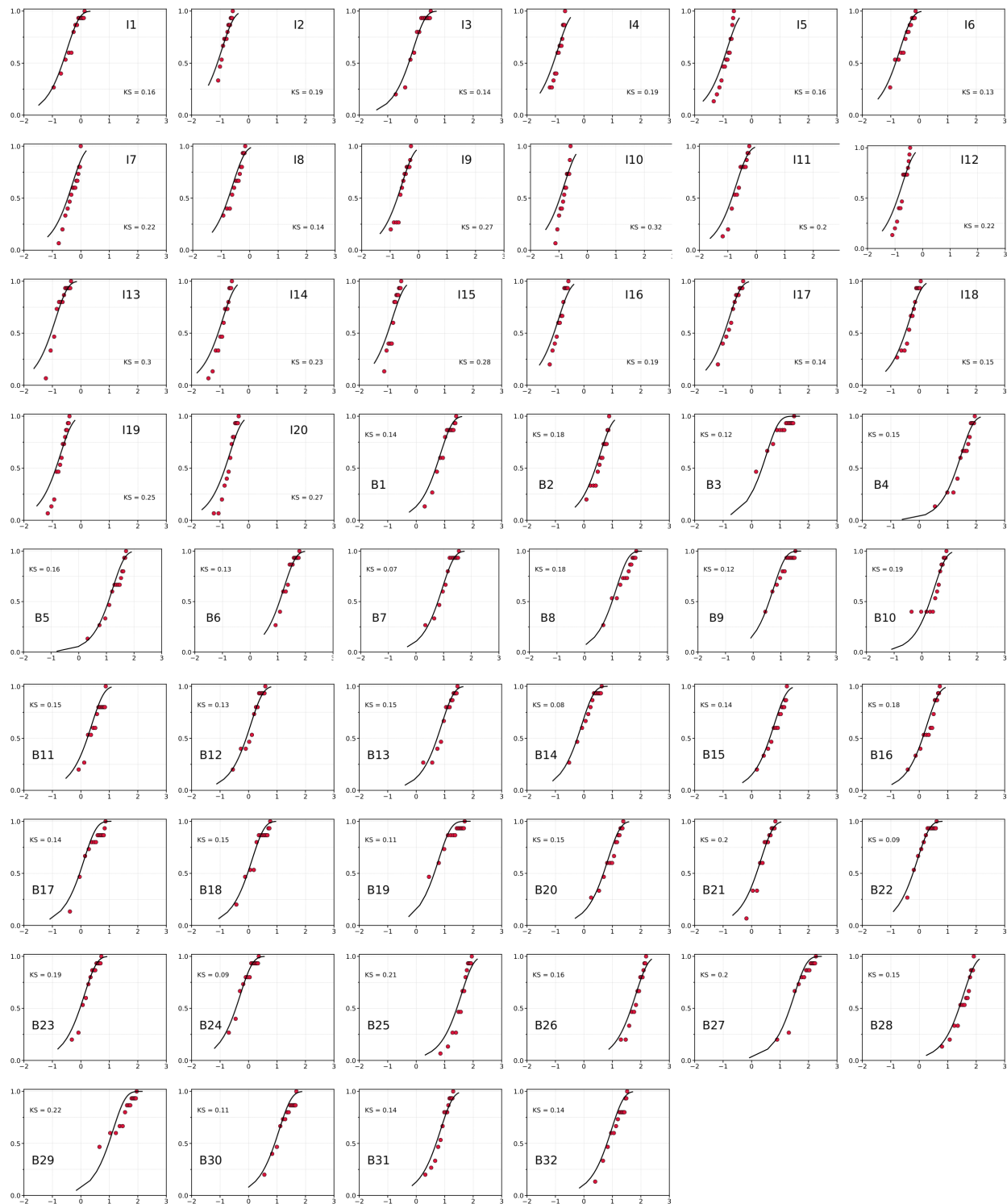
... intermediate state based on its different orientations observed in independent binding attempts, but which lies in a broader definition of intermediate state. Further classification is not consistent with lag time and meaningless. (B) The MSM derived metastable state populations. Also provided are estimated populations for unbound and bound states, based on the binding and unbinding times measured by MSM and infrequent metadynamics respectively. (C) The number of transitions observed between metastable states for a 3 state MSM.

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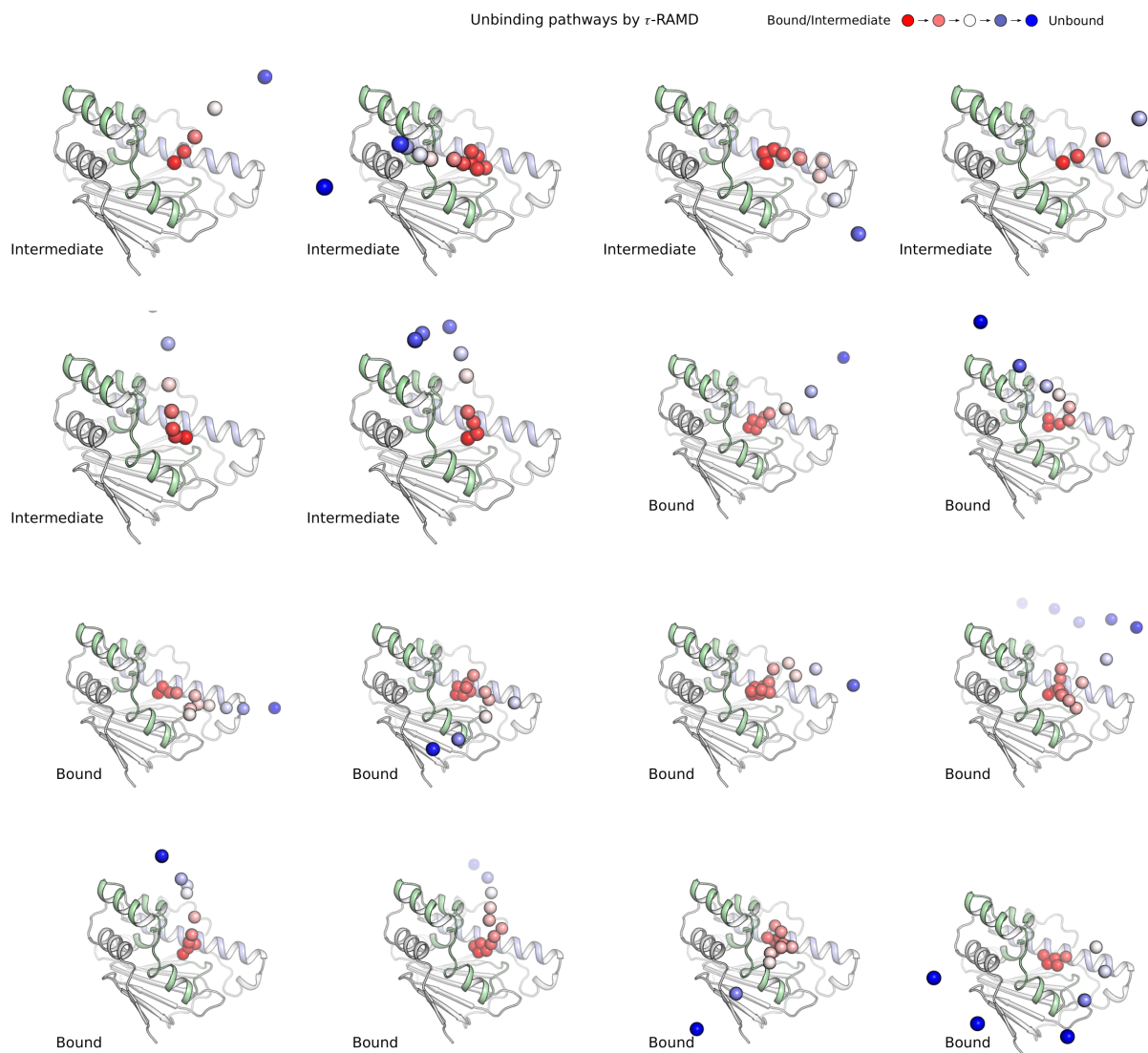


**Figure S3:** (A) The raw unbiased binding times estimated from 80 replicates of infrequent metadynamic simulations. (B) The empirical cumulative distribution function (ECDF) of unbiased unbinding times by fitting to poisson distribution. The fitted  $\tau$  and KS statistic of fitting is shown. (C) The representative GDM unbinding pathways via infrequent metadynamics. The unbinding occurs in all possible paths, with no specific preference.

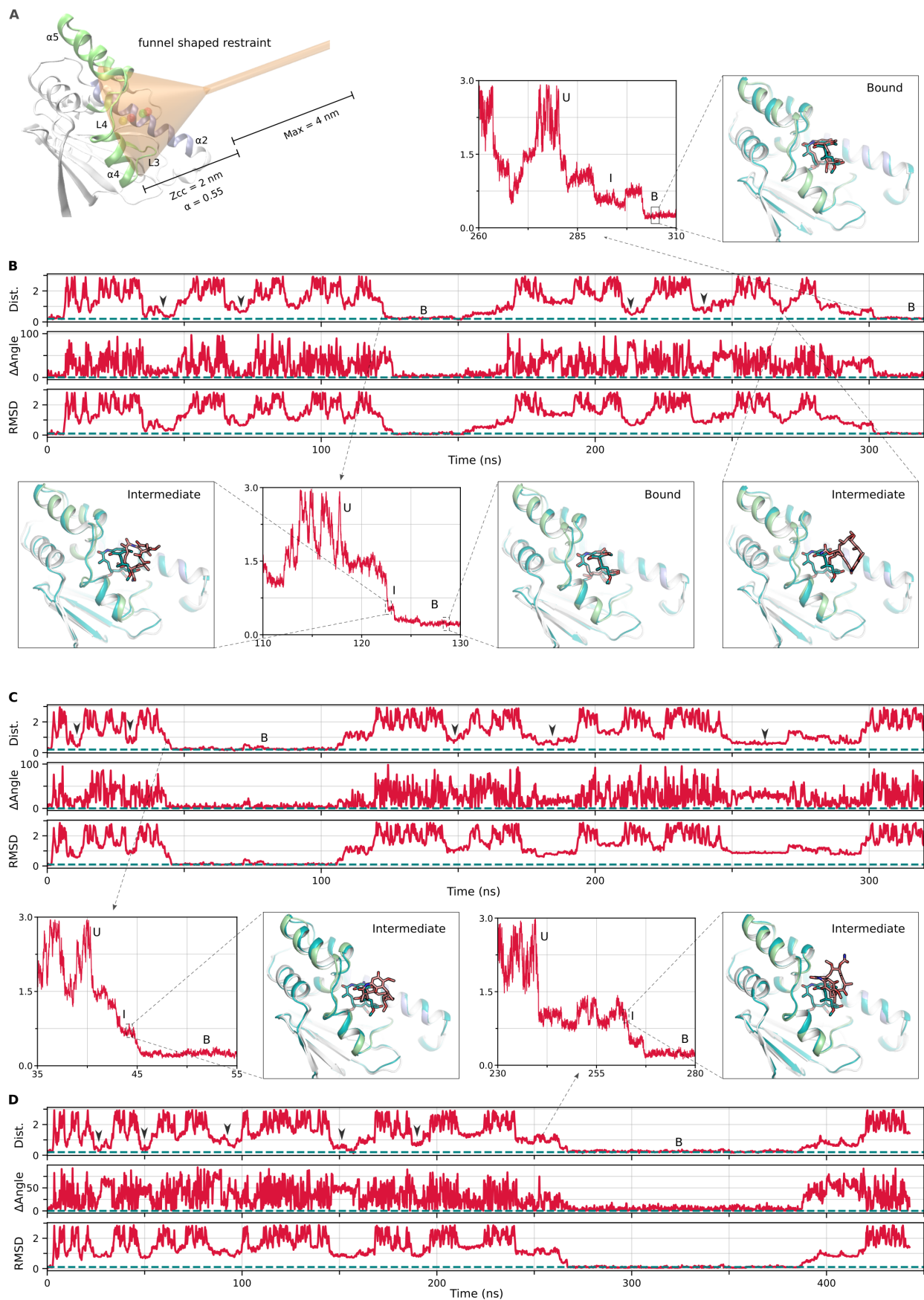




**Figure S4:** The raw statistics of residence times measured via  $\tau$ -RAMD. The I1-I20 and B1-B32 represents the intermediate and bound residence times respectively. The x and y axis represents the log10 and empirical cumulative distribution functions of 15 residence times measured by independent replicates.

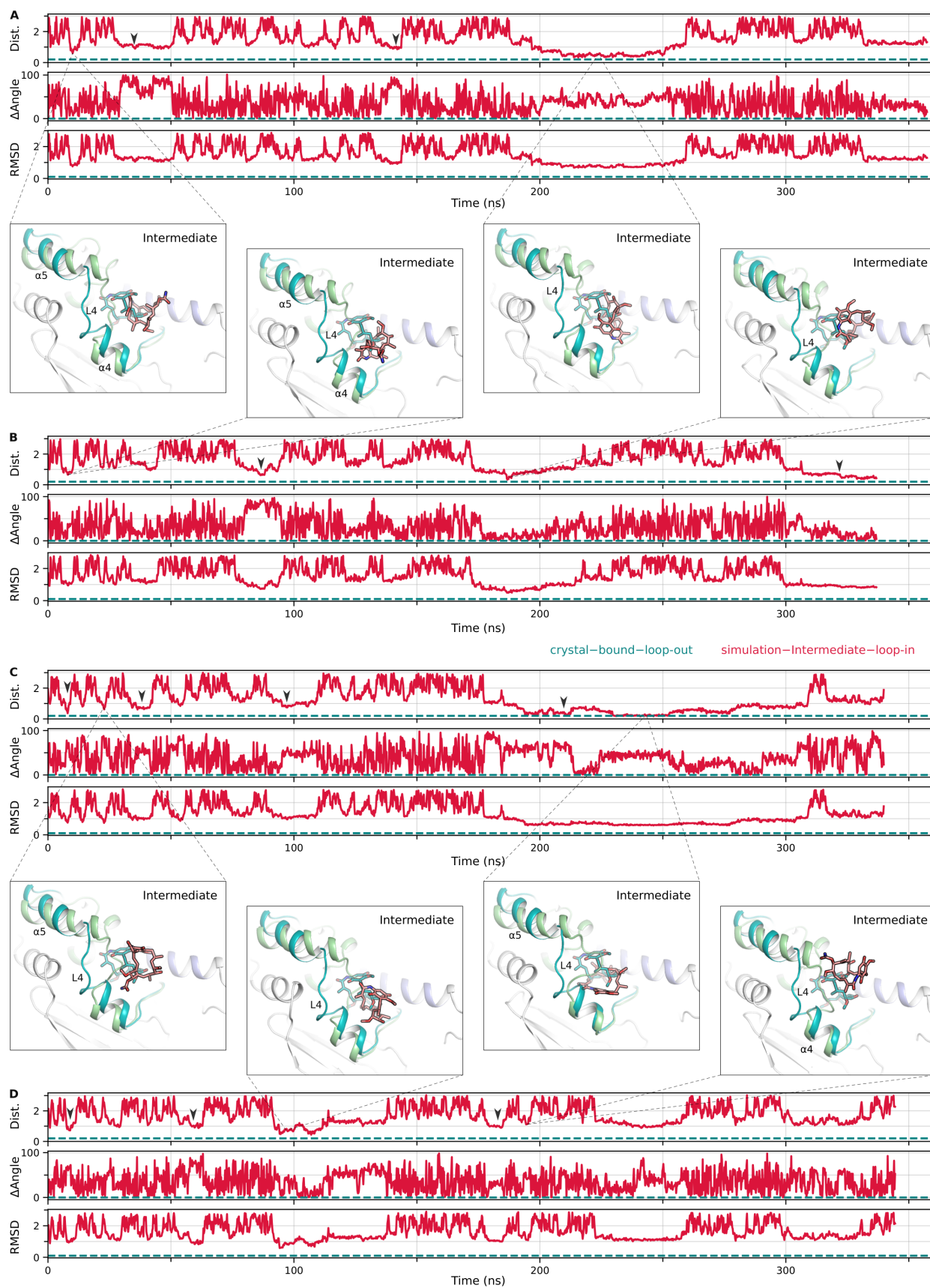


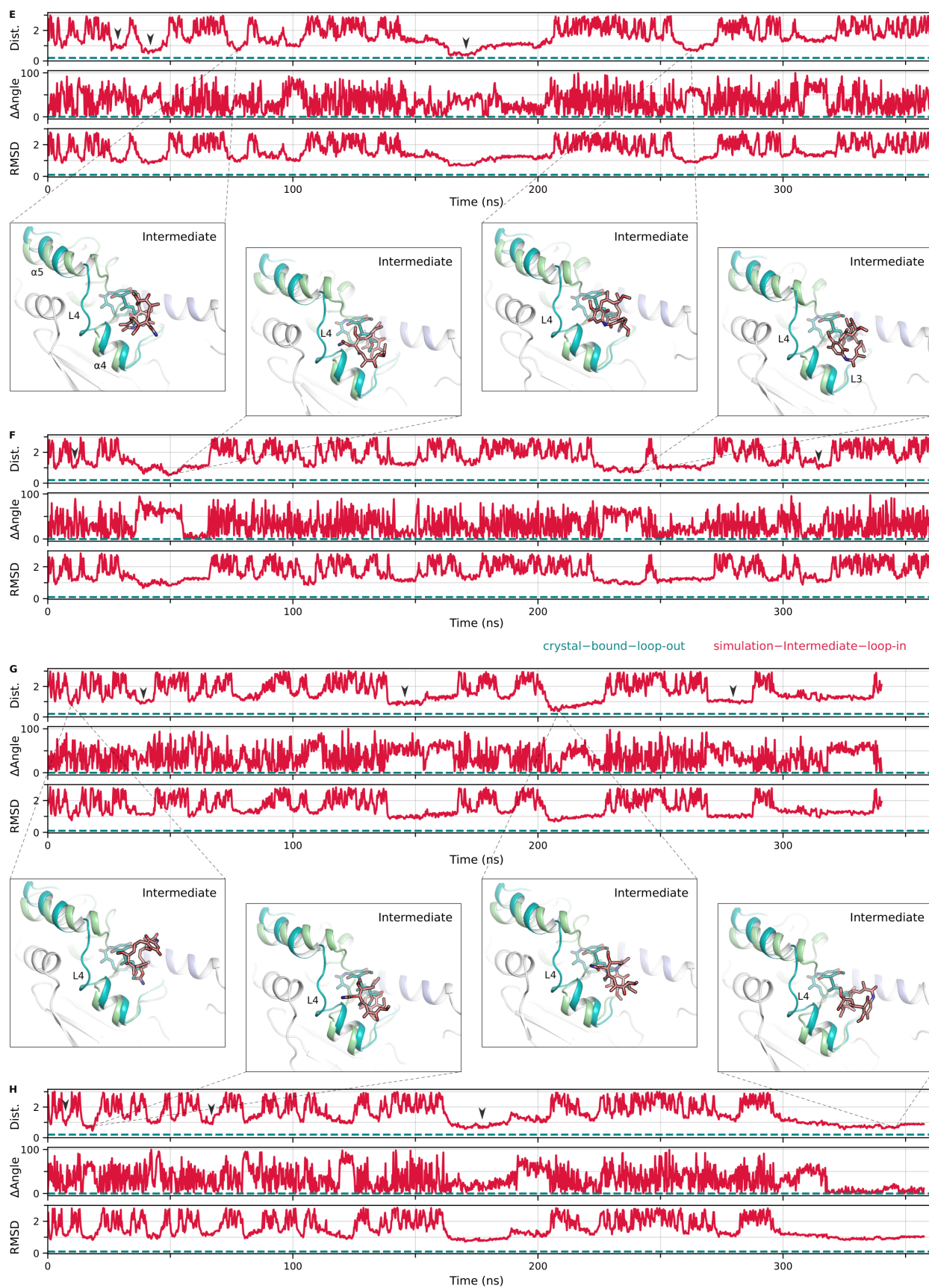
**Figure S5:** The representative unbinding paths by  $\tau$ -RAMD. In agreement with infrequent meta-dynamics, the unbinding paths are not specific. All possible paths were observed.



**Figure S6:** (A) The funnel setup for funnel metadynamics simulations. The cone part of funnel sufficiently include the binding pocket including L3,  $\alpha$ 4, L4, and parts of  $\alpha$ 2,  $\alpha$ 5 and  $\alpha$ 6 helices. (B-D) The distance,  $\Delta$ angle and RMSD of GDM during funnel-metadynamics simulations. The arrowhead highlights the various intermediate states achieved during unsuccessful binding attempts. The zoomed plots highlighting the three state binding mechanism are shown along with the structural snapshots, compared to crystal bound pose (teal color).

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**Figure S7:** (A-H) The distance,  $\Delta$ angle and RMSD of GDM during the funnel metadynamics simulations started from loop-in conformation of L4. The dashed teal line represents the values in bound state. No successful binding were observed, in agreement with unbiased trajectories where intermediate state was achieved in loop-in conformation which could not transition to bound state (Fig. 2A). Snapshots for some of the intermediate states were shown, overlayed with the bound state in loop-out conformation (teal color). Some other intermediate states were marked by arrow-heads.

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funnel setup used in this work for excited states (detailed in methods). (C) The distance (nm),  $\Delta$ Angle, and RMSD (nm) of three funnel metadynamics replicates whereby the ligand is able to reach the intermediate-like or bound-like states. (D) The snapshots from funnel metadynamics, overlaid with crystal bound state. The snapshots of ‘D’ are taken from marked squares from subfigure ‘C’. (E) The GDM binding energy profile in excited conformation, averaged from the three funnel replicates of ‘C’. The shaded portion represents the error bar. (F) The raw unbiased binding times of inhibitor GDM bound in excited conformation, via infrequent metadynamics. (G) The representative GDM unbinding pathways in excited conformations via infrequent metadynamics. The value in brackets represents the number of similar pathways.

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## Supplementary results - I: Evaluating the inhibitor binding and unbinding in excited conformation of N-Hsp90

Prior to 2020, to best of our knowledge, all the crystal structures of Hsp90 were determined for its N terminal domain, whereby the L4 was determined in loop-in, loop, loop-out and helical conformations (Fig. 1). The recent determination of full length structures, exhibit yet another conformation of N-Hsp90, called as ‘excited’ or ‘lid-closed’ etc, whereby the entire  $\alpha 5$  and unfolded  $\alpha 6$  moved significantly to cover the binding pocket (Fig. 1).<sup>33</sup> This the determinants of this conformational remained a matter of research,<sup>34–36</sup> nucleotide binding was found enough to transition to this conformation.<sup>33</sup> But it was considered, that this conformation might be at play in full length Hsp90, where the  $\alpha 1$  interchange between the dimers for the excited conformation to be achieved.

However, recent crystal structure of only N-Hsp90 bound with non-GDM inhibitor was determined in excited conformation (PDB-7UR3).<sup>37</sup> Very recent NMR investigation,<sup>21</sup> identified that the excited conformation can be sampled in N-Hsp90 within apo state, opening new avenues of research. Though it was identified to be a minor state (3.2%), with very high kinetic barrier (approximately 12 ms). Furthermore, unlike the ground state conformations

which are specific for inhibitors, the excited state conformation seems to be a final bound conformation for all the ligands that is achieved in full Hsp90. Hence, we considered the binding and unbinding potential of this conformation (Fig. S8).

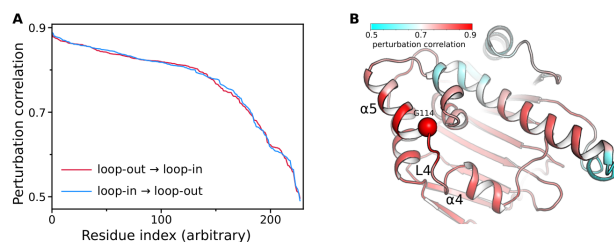
We considered two conformations, within broader definition of excited conformation, to study the inhibitor binding mechanism (Fig. S8A). The binding pocket is completely covered in this conformation, and the GDM binding mode is in steric clashes with unfolded  $\alpha 6$  (Fig. S8A), hence binding to this conformation might not be feasible. Nevertheless, we attempted to model binding mechanism, as ligands are known to bound the hidden, buried or closed pockets.<sup>38,39</sup> We used similar funnel setup as for ground state conformation (Fig. S8B). The excited conformation of PDB 8B7J is shifted towards the  $\alpha 2$ , more hindering the binding path in the direction of funnel, hence we considered an additional funnel setup for this excited conformation, in-between  $\alpha 4$ , L4, and  $\alpha 5$  (Fig. S8B). In total, we performed eight replicates of binding simulations via funnel metadynamics for each excited conformation (7UR3:8, 8B7J:4+4).

In none of the eight replicates of excited-8B7J conformation, the inhibitor was able to reach close to binding pocket, neither intermediate like, nor bound-like. As a contrast, in ground state funnel metadynamics, the GDM achieves intermediate state multiple times in each replicate (Fig. 2K, S6, S7). In eight replicates of excited-7UR3, the GDM was able to reach the binding pocket in three replicates (Fig. S8C). In Traj-1, the GDM was able to achieve a bound-like state, within the binding pocket but different binding mode compared to crystal-1YET of GDM (Fig. S8D). Maybe that GDM might have a different binding mode in excited conformation. However, this observation was not reproducible in other replicates, whereby GDM was able to reach the intermediate-like states (Fig. S8C,D). Furthermore, the overall binding energy in excited conformation comes out to be positive (Fig. S8E), indicating that binding in this conformation is not feasible.

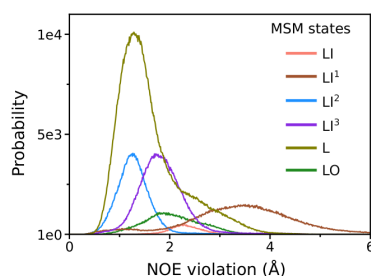
We also estimated the unbinding kinetics of GDM in the excited conformation. Since, this conformation completely covers the binding pocket, the unbinding from this conformation

also seems unlikely. Since, the direct positioning of GDM in excited conformation results in steric overlaps, and bound mode was not achieved in funnel metadynamics, the GDM bound excited conformation was achieved via targeted simulations (methods). We started with GDM bound ground state (loop-out) conformation, and performed the targeted simulations to achieve the excited conformations of either type 7UR3 or 8B7J. The GDM bound excited conformations were achieved within 1-3 Å in 110-130 rounds of targeted simulations. We performed 60 replicates of infrequent metadynamics on GDM bound in excited conformation (30 from each excited conformation). Though with constant deposition of bias for long time, the GDM was finally able to unbind from the excited conformation, residence time calculations indicate the time of the order of  $10^{20} - 10^{30}$  seconds, which is impossible (Fig. S8F). The unbinding paths in excited conformation seems to be specific, but these are not feasible in real time. Hence, we concluded that unbinding from the excited conformation is not feasible.

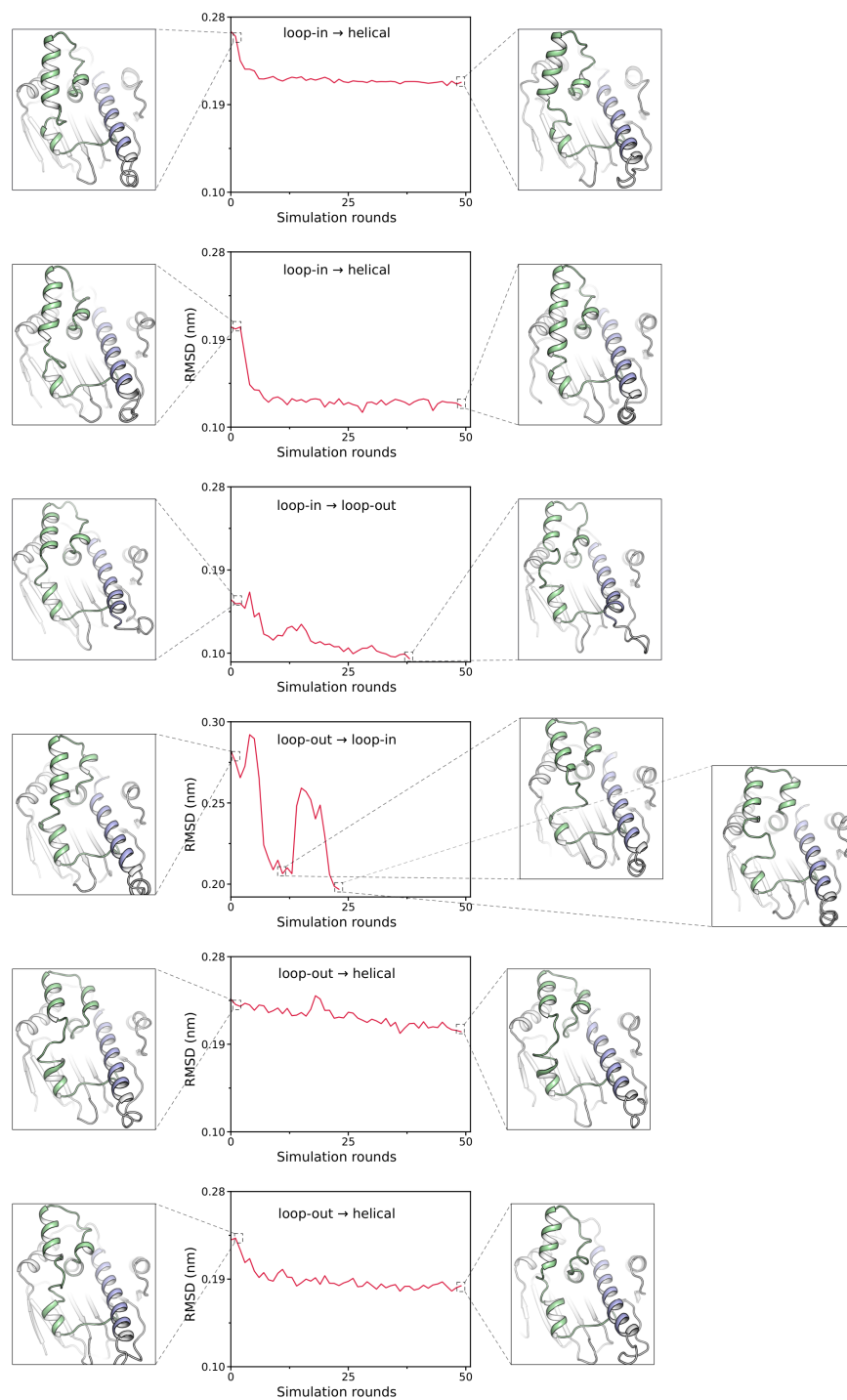
Overall, we concluded that though this state is very crucial, specifically in chaperoning cycle,<sup>33</sup> and is accessible in apo state,<sup>21</sup> this conformation is not involved in inhibitor binding or unbinding. These results are not in disagreement with the works of Honet et al.<sup>21</sup>



**Figure S9:** (A,B) Perturbation correlation estimated via PRS analysis for different transitions, and overlayed on N-Hsp90 structure as color-code. Sphere represent G114 with highest correlation. Arbitrary residue index correspond to correlation values in decreasing order. In PRS, starting from N-Hsp90 in one conformation (e.g., loop-out), small random perturbations of force and direction were applied to each residue to determine whether the perturbations could drive N-Hsp90 into the loop-in conformation. If perturbations applied to most residues lead to the loop-out to loop-in transition, it suggests that the transition is spontaneous, and vice versa. PRS was performed for both *loop-in*  $\rightarrow$  *loop-out* and *loop-out*  $\rightarrow$  *loop-in* transitions, using the crystallographic poses of the loop-out (1YES) and loop-in (1YER) conformations of the apo form as reference end states. The perturbation correlation analysis indicated that all residues near the binding pocket exhibited similar and high correlations ( $>0.7$ ). Thus, we concluded that the loop-in and loop-out conformations are spontaneously interconvertible, consistent with MSM timescales.



**Figure S10:** The NMR NOE distance violations measured in apo simulations, compared to experimental values.<sup>21</sup> The calculations were performed for each metastable state of apo MSM.



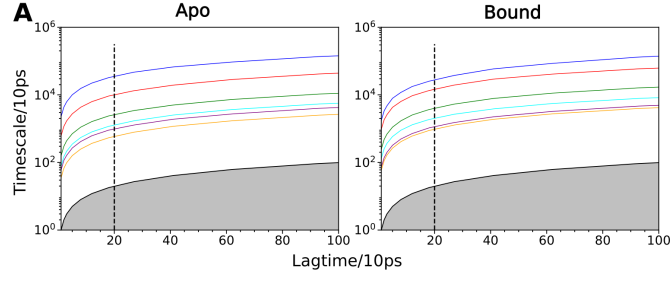
**Figure S11:** The FAST adaptive sampling simulations for different conformational transitions of L4, and associated snapshots of initial and final rounds. The RMSD is compared to the target conformation (PDBs are helical:9AUU,8AGI, loop-in:1YES, loop-out:1YER).<sup>40,41</sup> Though the *loop-out* → *loop-in* conformation was achieved closed to 1 Å, but a loop-in conformation was easily achieved, as depicted in its snapshot.

## Supplementary results - II: Evaluating the sampling of helical conformation within apo state

The ‘helical’ conformation of loop L4 was not sampled in any of our unbiased simulations, despite its structural proximity to the loop-in, loop and loop-out ground state conformations (Fig. 1). Also, the the helical conformation was sampled very closely in simulations ( $<2$  Å RMSD), as loop metastable state (loop conformation, L) of apo MSM, yet it is not perfectly achieved in its helical form in the simulations. To examine whether this was due to limitations in sampling, we employed targeted adaptive sampling using the FAST algorithm<sup>42</sup> to drive unbiased transitions toward the helical state. This approach involves iterative rounds of MD simulations guided by structural similarity to a defined target (methods). For example, simulations initiated from the loop-in conformation readily transitioned to the loop-out state within 25 rounds (Fig. S9, S11). However, attempts to drive the loop-in conformation toward the helical state (as observed in PDB ID: 9AUU) consistently failed, suggesting that the helical state is entropically disfavored in the apo ensemble.

We further validated this result by attempting transitions from alternative starting points, namely, the loop-out state to the helical conformation (PDB IDs: 9AUU and 8AGI). In both cases, the helical state could not be reached even after 50 rounds of simulation. In contrast, loop-out to loop-in transitions were achieved within 10 rounds (Fig. S11).

Taken together, these findings indicate that while L4 can spontaneously interconvert between loop-in, loop-out, and intermediate loop states in the apo form, transitions to the helical conformation are not observed. This suggests a high entropic penalty associated with helix formation in the absence of stabilizing ligand interactions. Consistent with this, all experimentally resolved helical L4 conformations have been observed exclusively in the ligand-bound state, where direct contacts with the inhibitor likely stabilize the helical structure.



|                 | LI         | LI <sup>2</sup> | LO         | LI <sup>1</sup> | LI <sup>3</sup> | L          |
|-----------------|------------|-----------------|------------|-----------------|-----------------|------------|
| LI              | 0          | 7.55±0.21       | 9.08±0.06  | 12.7±1.10       | 20.88±3.09      | 4.58±0.03  |
| LI <sup>2</sup> | 13.61±0.10 | 0               | 15.04±0.12 | 7.20±1.74       | 10.47±2.77      | 10.6±0.10  |
| LO              | 13.80±0.08 | 13.38±0.17      | 0          | 20.18±1.32      | 27.12±3.04      | 0.19±0.00  |
| LI <sup>1</sup> | 12.67±0.15 | 3.26±0.55       | 15.61±0.09 | 0               | 14.57±3.46      | 11.16±0.07 |
| LI <sup>3</sup> | 16.42±0.65 | 2.06±0.59       | 17.93±0.50 | 9.51±2.28       | 0               | 13.49±0.52 |
| L               | 13.29±0.08 | 12.88±0.17      | 3.54±0.03  | 19.68±1.32      | 26.62±3.04      | 0          |

|                | L         | L <sup>1</sup> | E'         | LO        |
|----------------|-----------|----------------|------------|-----------|
| L              | 0         | 21.2±0.18      | 13.43±1.62 | 1.38±0.05 |
| L <sup>1</sup> | 4.25±0.00 | 0              | 17.46±1.62 | 5.27±0.09 |
| E'             | 9.36±0.50 | 32.1±0.60      | 0          | 3.29±0.78 |
| LO             | 3.48±0.11 | 26.24±0.28     | 10.53±1.50 | 0         |

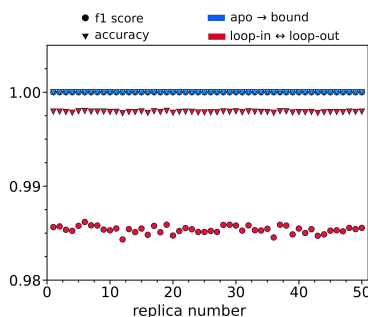
  

|  | LI        | LI <sup>2</sup> | LO       | LI <sup>1</sup> | LI <sup>3</sup> | L          |
|--|-----------|-----------------|----------|-----------------|-----------------|------------|
|  | 1.51±0.01 | 11.92±0.17      | 6.4±0.02 | 14.53±0.17      | 16.3±0.32       | 49.34±0.40 |

|  | L          | L <sup>1</sup> | E'        | LO         |
|--|------------|----------------|-----------|------------|
|  | 12.91±0.39 | 15.81±0.12     | 23.9±1.47 | 47.38±1.57 |

**Figure S12:** (A) The ITS plots for MSM models of apo and bound states. (B,C) Mean first passage transition times ( $\mu s$ ) and equilibrium populations of different states of apo and bound MSMs.



**Figure S13:** The accuracy and f1 scores for different replicas of trained RF on test data.

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